

Cell-State Reachability

MEASURED → MODELLED → TEST NEXT

A screen-relative test of transcriptomic direction — not a state-conversion claim

1

Inputs

source objects reused from Zhu workflow

TARGET DIRECTION **d**

External Th1-vs-Th2 population contrast
(not an observed trajectory)

EFFECT DICTIONARY **E**

Screen-derived CRISPRi DE z-scores
in post-expansion Rest CD4 cells

Raw screen overlap: 9,831 / 25,672

Final merged analysis: 6,188 genes

Dictionary is donor-collapsed; not
measured in polarized Th2 cells

2

Projection

what the model computes

target

unmatched residual

projected model fit

Non-negative linear combinations
in screen z-score space
Residual is model-relative

3

Challenge

frozen retrospective diagnostics

0.446 ± 0.010

mean ± SD, 12 fixed random-gene splits

external Th1-like direction

historical fixed split: 0.448

60/60 diagnostic target shuffles
below observed

plus-one empirical p = 1/61

Retrospective directional evidence;
not donor or functional validation

4

Decide

what this result changes

REPLICATION NEEDED

Donor-resolved, polarized
Th2 starting-state study

MEASURE NEXT

Matched modalities, combinations
+ functional readouts

NOT ESTABLISHED

State conversion, rescue,
or target validation

CLAIM BOUNDARY

Screen-relative directional support • measured subset only • model-relative geometry ≠ biological efficacy